Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the validation accuracy during training.

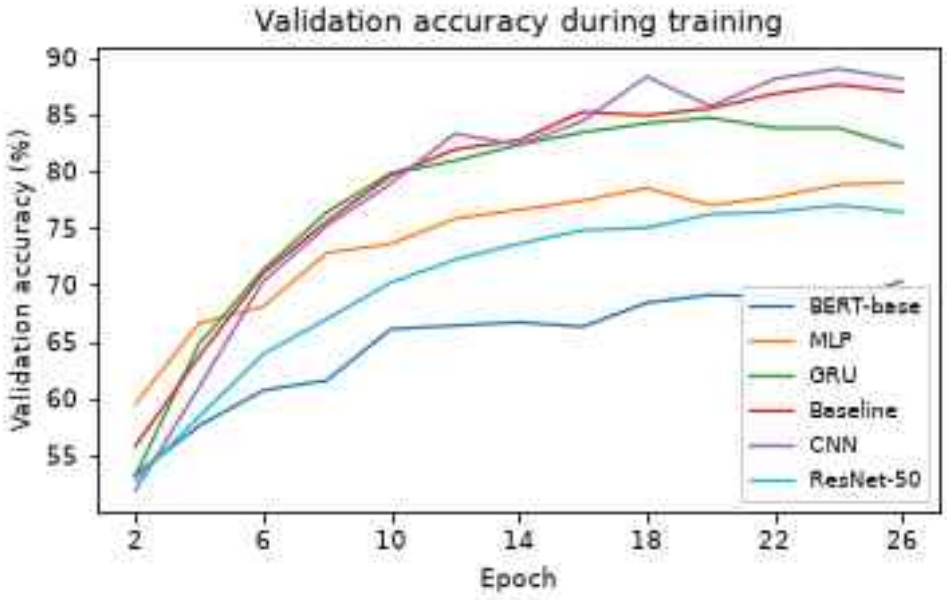


Figure 1: Validation accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 10, BERT-base achieves 66.1% validation accuracy. As shown in Figure 1, GRU reaches 75.6% validation accuracy at epoch 24.

Figure 2 shows the validation accuracy during training.

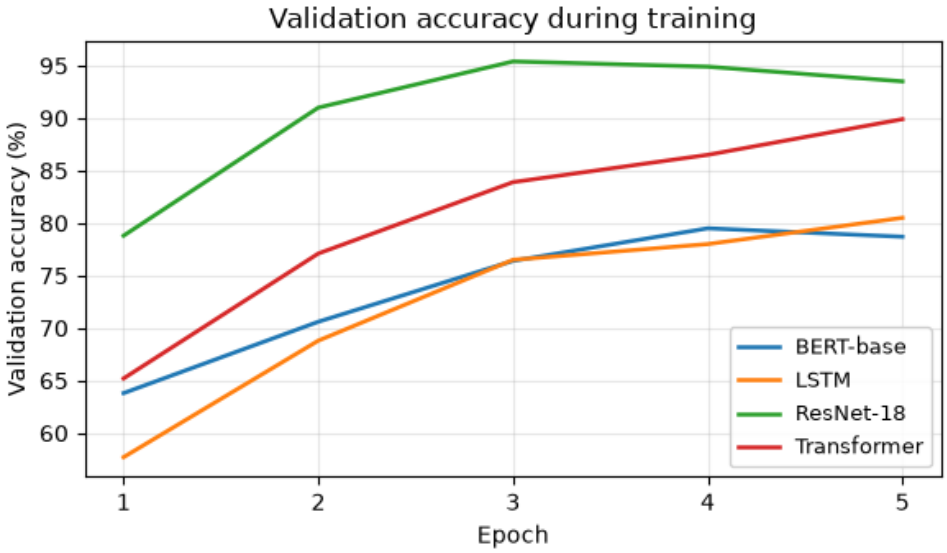


Figure 2: Validation accuracy during training for the models in this experiment.

As shown in Figure 2, BERT-base attains a validation accuracy of 78.7% at epoch 5. From epoch 1 to epoch 4, the validation accuracy of LSTM improves by a relative 35%.

3 Discussion

On AG News, XGBoost obtains an accuracy of 63.0%. Overall, the results are in line with what we expected from the literature.